Rates of Changes

Consider a running car,

Let t be the time (in second)
s be the displacement (in metre)
v be the velocity
a be the acceleration



then

$$\text{Velocity} = \frac{\text{change of displacement } s}{\text{time taken}} \qquad v = \frac{ds}{dt} \quad (\text{ms}^{-1})$$

$$\text{Acceleration} = \frac{\text{change of velocity } v}{\text{time taken}} \qquad a = \frac{dv}{dt} \quad (\text{ms}^{-2})$$

Velocity and acceleration are examples of rates.

Here are more examples:



Grow at
0.1 cms^{-1}

Rate of changes of the height h

$$\frac{dh}{dt} = 0.1 \text{ cms}^{-1}$$

(+ve means increasing)



Air leaks at
2 cm^{3}s^{-1}

Rate of changes of the volume V

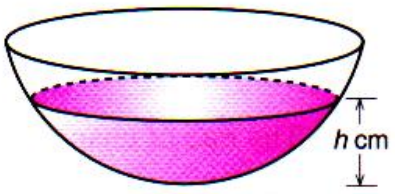
$$\frac{dV}{dt} = -2 \text{ cm}^3\text{s}^{-1}$$

(–ve means decreasing)

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Rates of Changes

e.g. Water is being leaked out at a rate of π cm^3s^{-1} from a vessel. Volume of water is $V = \frac{\pi}{3}(30h^2 - h^3)$, where h is the depth of the water in cm. Find the rate of change of the water level (水位變化的速度).



Step 1 - Write down the required rate in terms of derivative

The rate of change of the water level

$$= \frac{\text{change of water level } h}{\text{time taken}} = \frac{dh}{dt}$$

Note!!!

$\because$ V is decreasing

$\therefore$ $\frac{dV}{dt}$ is –ve !

Step 2 - Write down the given rate(s) of changes

Water leaks at **π cm^3s^{-1}**

$$\Rightarrow \quad \frac{\text{change of water volume } V}{\text{time taken}} = \frac{-\pi \text{ cm}^3}{1 \text{ s}} \quad \Rightarrow \quad \frac{dV}{dt} = -\pi \text{ cm}^3\text{s}^{-1}$$

Step 3 - Differentiate the given function:

$$V = \frac{\pi}{3}(30h^2 - h^3) \qquad \therefore \frac{dV}{dh} = \frac{\pi}{3}(60h - 3h^2) \quad = \pi(20h - h^2)$$

Final Step - Apply Chain Rule to connect all the rates

$$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$$

$$-\pi = \pi(20h - h^2) \cdot \frac{dh}{dt}$$

$$\frac{dh}{dt} = -\frac{1}{20h - h^2} \text{ cm s}^{-1}$$

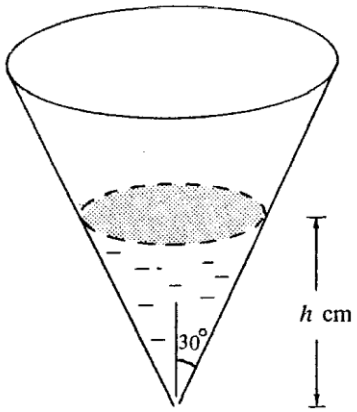
Use **chain rule** to connect $\frac{dh}{dt}$, $\frac{dV}{dt}$ and $\frac{dV}{dh}$

$\therefore$ Rate of change of water level is $-\frac{1}{20h - h^2}$ cms^{-1}.

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HKCEE Questions (level 2)

1. ***[A.Math 1992 Paper I Q.7]***



The figure shows a vessel in the shape of a right circular cone with semi-vertical angle 30°. Water is flowing out the cone through its apex at a constant rate of π cm^3s^{-1}.

(a) Let V cm3 be the volume of water in the vessel when the depth of water is h cm. Express V in terms of h.

(b) How fast is the water level falling when the depth of water is 4 cm?

(7 marks)

Not for sale

HKDSE Questions (level 2)

2. ***[M2 2017 Q.6]***

A container in the form of an inverted right circular cone is held vertically. The height and the base radius of the container are 20 cm and 15 cm respectively. Water is now poured into the container.

(a) Let A cm^2 be the wet curved surface area of the container and h cm be the depth of water in the container. Prove that $A = \frac{15}{16}\pi h^2$.

(b) The depth of water in the container increases at a constant rate of $\frac{3}{\pi}$ cm/s. Find the rate of change of the wet curved surface area of the container when the volume of water in the container is 96π cm^3. ***(6marks)***

Not for sale

HKDSE Questions (level 2)

3. ***[M2 2019 Q.3]***

A researcher performs an experiment to study the rate of change of the volume of liquid X in a vessel. The experiment lasts for 24 hours. At the start of the experiment, the vessel contains $580\,\text{cm}^3$ of liquid X. The researcher finds that during the experiment, $\frac{\mathrm{d}V}{\mathrm{d}t} = -2t$, where $V\,\text{cm}^3$ is the volume of liquid X in the vessel and t is the number of hours elapsed since the start of the experiment.

(a) The researcher claims that the vessel contains some liquid X at the end of the experiment. Is the claim correct? Explain your answer.

(b) It is given that $V = h^2 + 24h$, where $h\,\text{cm}$ is the depth of liquid X in the vessel. Find the value of $\frac{\mathrm{d}h}{\mathrm{d}t}$ when $t = 18$.

(6 marks)

Not for sale

HKDSE Questions (level 2)

4. ***[M2 2012 Q.6]***

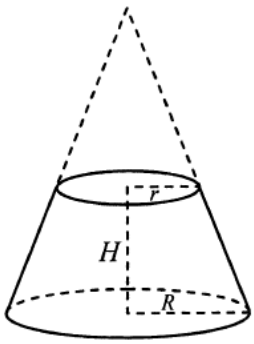


Figure 1

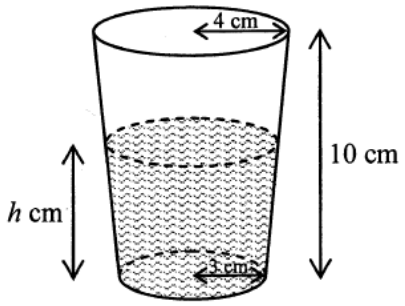


Figure 2

A frustum of height H is made by cutting off a right circular cone of base radius r from a right circular cone of base radius R (see Figure 1). It is given that the volume of the frustum is $\frac{\pi}{3}H(r^2+rR+R^2)$.

An empty glass is in the form of an inverted frustum described above with height 10 cm, the radii of the rim and the base 4 cm and 3 cm respectively. Water is being poured into the glass. Let h cm $(0 \le h \le 10)$ be the depth of the water inside the glass at time t s (see Figure 2).

(a) Show that the volume V cm^3 of water inside the glass at time t s is given by

$$V = \frac{\pi}{300}(h^3+90h^2+2700h).$$

(b) If the volume of water in the glass is increasing at the rate 7π cm^{3}s^{-1}, find the rate of increase of depth of water at the instant when $h = 5$. ***(6 marks)***

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HKCEE Questions (level 2)

5. ***[A.Math 2007 Q.9]***

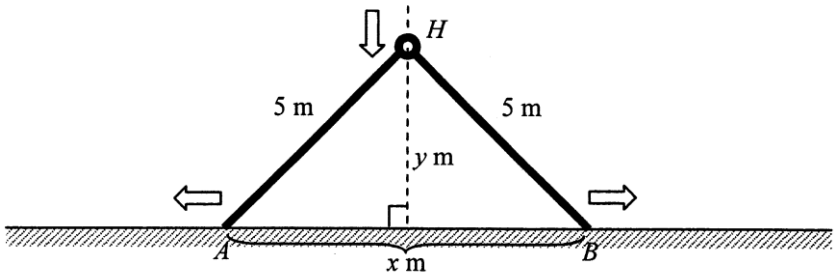


Figure 3

Two rods HA and HB, each of length 5 m, are hinged at H. The rods slide such that A, B and H are on the same vertical plane and A, B move on opposite directions on the horizontal floor, as shown in Figure 3. Let AB be x m and the distance of H from the floor be y m.

(a) Write down an equation connecting x and y.

(b) When H is 3 m from the ground, its falling speed is $2ms^{-1}$. Find the rate of change of distance between A and B with respect to time at that moment.

(5 marks)

Not for sale

HKCEE Questions (level 2)

6. *[A.Math 1997 Paper I Q.4]*

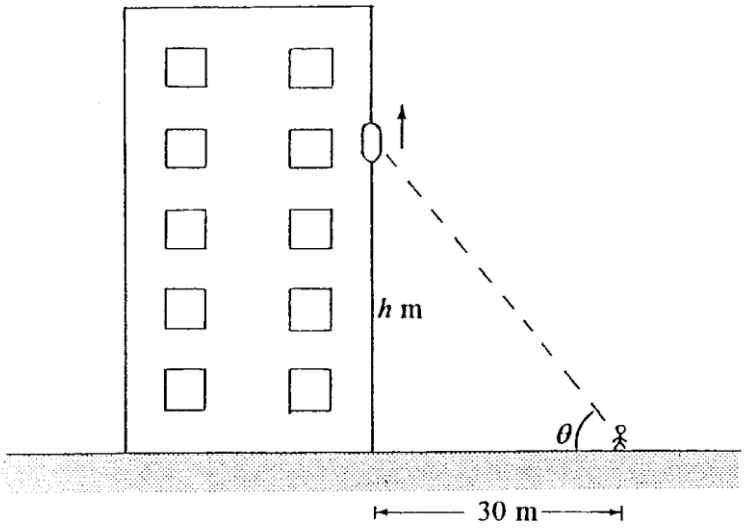


Figure 1

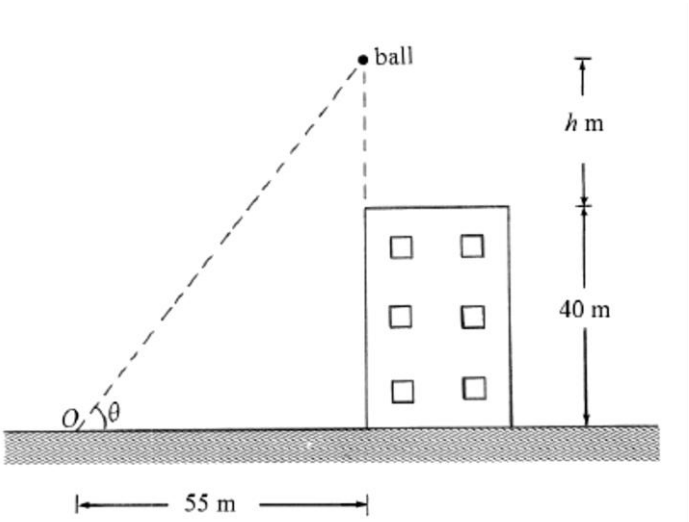
A man stands at a horizontal distance of 30 m from a sight-seeing elevator of a building as shown in Figure 1. The elevator is rising vertically with a uniform speed of 1.5 m s^{-1}. When the elevator is at a height h m above the ground, its angle of elevation from the man is θ. Find the rate of change of θ with respect to time when the elevator is at a height $30\sqrt{3}$ m above the ground. [Note : You may assume that the sizes of the elevator and the man are negligible.]

(5 marks)

Not for sale

HKCEE Questions (level 2)

7. ***[A.Math 1999 Paper I Q.8]***



A ball is thrown vertically upwards from the roof of a building 40 meters in height. After t seconds, the height of the ball above the roof is h meters, where $h = 20t - 5t^2$.
At this instant, the angle of elevation of the ball from the point O, which is at horizontal distance of 55 meters from the building is θ.

(a) **(i)** Find $\tan\theta$ in terms of t.

(ii) The value of θ when $t = 3$.

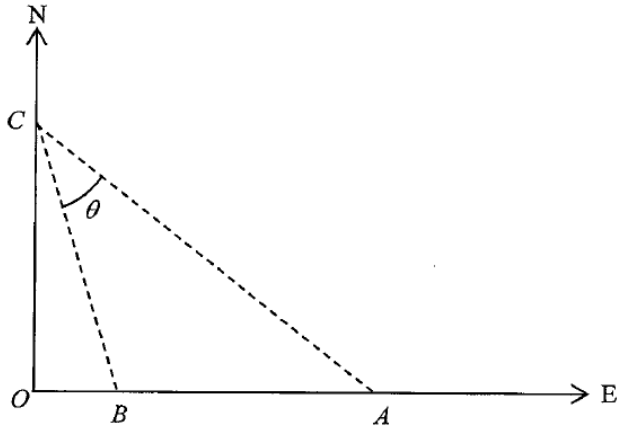
(b) Find the rate of change of θ with respect to time when $t = 3$. ***(7 marks)***

Not for sale

HKDSE Questions (level 2)

8. *[M2 2024 Q.9]*

Two toy cars, A and B, move along a straight line towards east with constant speeds 4 metres per second and 1 metre per second respectively. They start at the point O at the same time. The fixed point C is 10 metres due north of O. The following figure shows the positions of A and B after t seconds. Let $\angle ACB = \theta$ radians.



(a) Prove that $\tan\theta = \dfrac{15t}{2(t^2+25)}$.

(b) It is given that $\angle BAC = \angle ACB$ when $t = T$. Find

(i) T,

(ii) the rate of change of θ when $t = T$.

(7 marks)

Not for sale

HKDSE Questions (level 3)

9. ***[M2 2016 Q.3]***

Consider the curve $C: y = 2e^x$, where $x > 0$. It is given that P is a point lying on C. The horizontal line which passes through P cuts the y-axis at the point Q. Let O be the origin. Denote the x-coordinate of P by u.

(a) Express the area of ΔOPQ in term of u.

(b) If P moves along C such that OQ increases at a constant rate of 6 units per second, find the rate of change of the area of ΔOPQ when $u = 4$.

(5 marks)

Not for sale

HKDSE Questions (level 3)

10. ***[M2 2022 Q.7]***

Consider the curve $\Gamma: y = \ln(x + 2)$, where $x > 0$. Let P be a moving point on Γ with h as its x-coordinate. Denote the tangent to Γ at P by L and the area of the region bounded by Γ, L and the y-axis by A square units.

(a) Prove that $A = \dfrac{h^2 + 4h}{2h + 4} - 2\ln(h + 2) + 2\ln 2$.

(b) If $h = 3^{-t}$, where t is the time measured in seconds, find the rate of change of A when $t = 1$.

(8 marks)

Not for sale

HKDSE Questions (level 3)

11. ***[M2 2020 Q.6]***

Consider the curve $C_1: y = 2^{x-1}$, where $x > 0$. Denote the origin by O. Let $P(u, v)$ be a moving point on C_1 such that the area of the circle with OP as a diameter increases at a constant rate of 5π square units per second.

(a) Define $S = u^2 + v^2$. Does S increase at a constant rate? Explain your answer.

(b) Let C_2 be the curve $y = 2^x$, where $x > 0$. The vertical line passing through P cuts C_2 at the point Q. Find the rate of change of the area of ΔOPQ when $u = 2$.

(7 marks)

Not for sale

HKDSE Questions (level 3)

12. *[M2 2023 Q.6]*

(a) Let r be a positive constant. Figure 1 shows the shaded region which is inside the circle $x^2+(y-r)^2=r^2$ and below the horizontal line $y=h$, where $0 \le h \le 2r$. Prove that the volume of the solid of revolution generated by revolving the shaded region about the y-axis is $\frac{\pi}{3}h^2(3r-h)$.

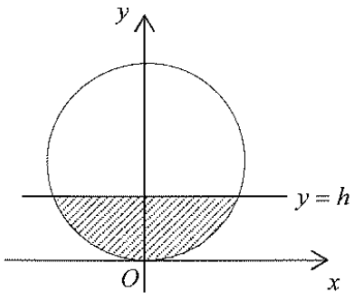


Figure 1

(b) A solid metal sphere of radius 10 cm is put into an empty right cylindrical container of radius 11 cm and height 20 cm, with longitudinal section as shown in Figure 2. Starting from time $t=0$, water is added to the container at a constant rate of $1\,\text{cm}^3$ per second. Let h cm be the depth of water after t seconds. By expressing $\frac{\mathrm{d}h}{\mathrm{d}t}$ in terms of h, find the greatest value of $\frac{\mathrm{d}h}{\mathrm{d}t}$.

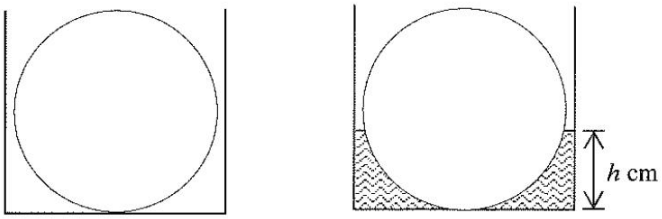


Figure 2

(7 marks)

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HKDSE Questions (level 3)

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HKCEE Questions (Level 3)

13. *[A.Math 1995 Paper I Q.9]*

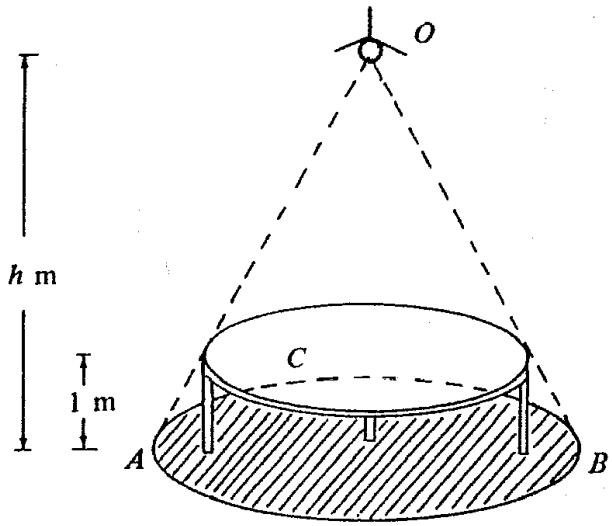


Figure 2

A small lamp O is placed h m above the ground, where $1 < h \leq 5$. Vertically below the lamp is the centre of a round table of radius 2 m and height 1 m. The lamp casts a shadow ABC of the table on the ground (see Figure 2). Let S m^2 be the area of the shadow.

(a) Show that $S = \dfrac{4\pi h^2}{(h-1)^2}$. (3 marks)

(b) If the lamp is lowered vertically at a constant rate of $\dfrac{1}{8}$ m s^{-1}, find the rate of change of S with respect to time when $h = 2$. (5 marks)

(c) Let V m^3 be the volume of the cone $OABC$.

(i) Show that $V = \dfrac{4\pi h^3}{3(h-1)^2}$.

(ii) Find the minimum value of V as h varies.

Does S attain a minimum when V attains its minimum? Explain your answer. (8 marks)

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HKCEE Questions (level 3)

Not for sale

HKDSE Questions (level 4)

14. *[M2 2014 Q.10]*

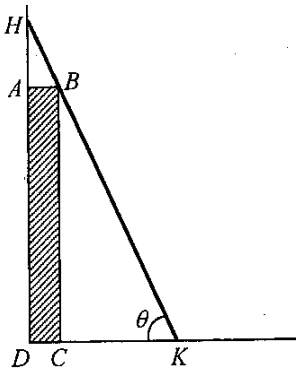


Figure 2

Thomas has a bookcase of dimensions $100cm \times 24cm \times 192cm$ at the corner in his room. He wants to hang a decoration on the wall above the bookcase. Therefore, he finds a ladder to climb up. Initially, the ladder touches the wall, the edge of the top of the bookcase and the floor at the same time, Let rectangle *ABCD* be the side-view of the bookcase and *HK* be the side-view of the ladder, so that *AB* = 24 cm and *BC* = 192 cm (see Figure 2). Let $\angle HKD = \theta$.

(a) Find the length of *HK* in terms of θ. ***(1 mark)***

(b) Prove that the shortest length of the ladder is $120\sqrt{5}$ cm. ***(5 marks)***

(c)

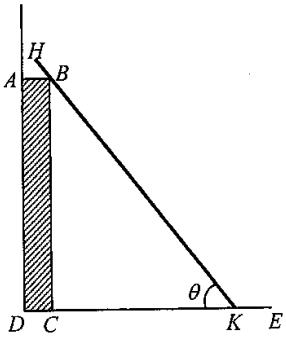


Figure 3

Suppose the length of the ladder is 270 cm. Suddenly, the ladder slides down so that the end of the ladder, *K*, moves towards *E* (see Figure 3). The ladder touches the edge of the top of the bookcase and the floor at the same time. Let *x* cm and *y* cm be the horizontal distances from *H* and *K* respectively to the wall.

(i) When *CK* = 160 cm, the rate of change of θ is $-0.1\,\text{rad}\,s^{-1}$. Find the rate of change of *x* at this moment, correct to 4 significant figures.

(ii) Thomas claims that *K* is moving towards *E* at a speed faster than the horizontal speed *H* is leaving the wall. Do you agree? Explain your answer.

(6 marks)

HKDSE Questions (level 4)

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HKDSE Questions (level 4)

15. ***[M2 2013 Q.12]***

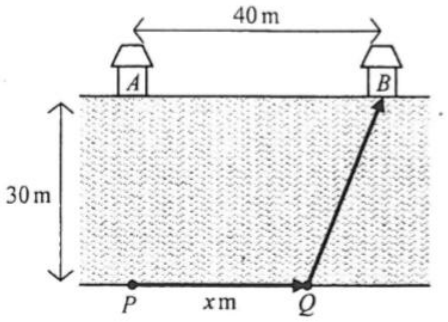


Figure 3

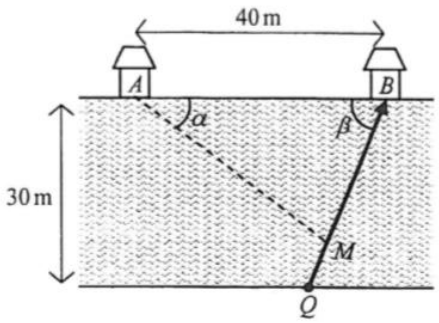


Figure 4

In Figure 3, the distance between two houses A and B lying on a straight river bank is 40 m. The width of the river is always 30 m. In the beginning, Mike stands at the starting point P in the opposite bank which is 30 m from A. Mike's wife, situated at A, is watching him running along the bank for x m at a constant speed of 7 ms^{-1} to point Q and then swimming at a constant speed of 1.4 ms^{-1} along a straight path to reach B.

(a) Let T seconds be the time that Mike travels from P to B.

(i) Express T in terms of x.

(ii) When T is minimum, show that x satisfies the equation $2x^2 - 160x + 2135 = 0$. Hence show that $QB = \dfrac{25\sqrt{6}}{2}$ m.

(6 marks)

(b) In Figure 4, Mike is swimming from Q to B with QB equals to the value mentioned in **(a)(ii)**. Let $\angle MAB = \alpha$ and $\angle ABM = \beta$, where M is the position of Mike.

(i) By finding $\sin\alpha$ and $\cos\beta$, show that $MB = \dfrac{200\tan\alpha}{\tan\alpha + 2\sqrt{6}}$.

(ii) Find the rate of change of α when $\alpha = 0.2$ radian. Correct your answer to 4 decimal places. ***(7 marks)***

Not for sale

HKDSE Questions (level 4)

Not for sale

HKCEE Questions (Level 4)

16. ***[2000 Paper I Q.13]***

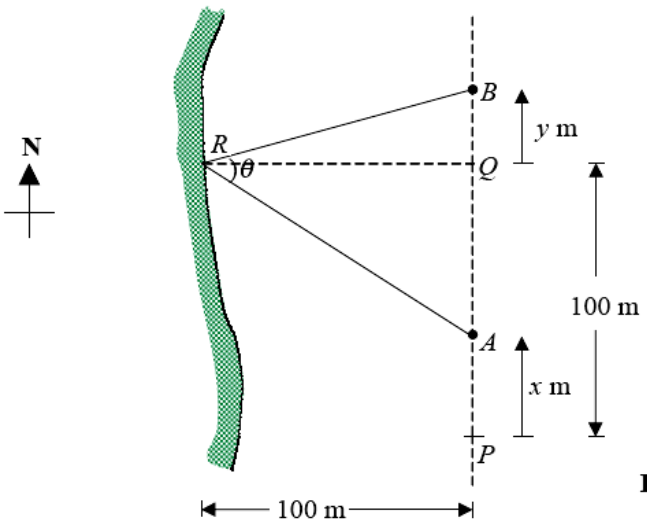


Figure 5

Two boats A and B are initially located at points P and Q in a lake respectively, where Q is at a distance 100 m due north of P. R is a point on the lakeside which is at a distance 100 m due west of Q. (See Figure 5.) Starting from time (in seconds) $t = 0$, boats A and B sail northwards. At time t, let the distances travelled by A and B be x m and y m respectively, where $0 \le x \le 100$. Let $\angle ARB = \theta$.

(a) Express $\tan \angle ARQ$ in terms of x.

Hence show that $\tan\theta = \dfrac{100(100 - x + y)}{10000 - 100y + xy}$. ***(4 marks)***

(b) Suppose boat A sails with a constant speed of 2 ms^{-1} and B adjusts its speed continuously so as to keep the value of $\angle ARB$ unchanged.

(i) Using (a), show that $y = \dfrac{100x}{200 - x}$.

(ii) Find the speed of boat B at $t = 40$.

(iii) Suppose the maximum speed of boat B is 3 ms^{-1}. Explain whether it is possible to keep value of $\angle ARB$ unchanged before boat A reaches Q.

(12 marks)

HKCEE Questions (level 4)

Not for sale

HKDSE Questions (level 4)

17. ***[M2 2018 Q.9]***

Consider the curve C: $y = \ln\sqrt{x}$, where $x > 1$. Let P be a moving point lying on C. The normal to C at P cuts the x-axis at the point Q while the vertical line passing through P cuts the x-axis at the point R.

(a) Denote the x-coordinate of P by r. Prove that the x-coordinate of Q is $\frac{4r^2 + \ln r}{4r}$.

(3 marks)

(b) Find the greatest area of ΔPQR. ***(5 marks)***

(c) Let O be the origin. It is given that OP increases at a rate not exceeding $32e^2$ units per minute. Someone claims that the area of ΔPQR increases at a rate lower than 2 square units per minute when the x-coordinate of P is e. Is the claim correct? Explain your answer. ***(4 marks)***

HKDSE Questions (level 4)

Not for sale

HKDSE Questions (level 4)

18. ***[M2 2021 Q.10]***

Denote the graph of $y=\sqrt{x^2+36}$ and the graph of $y=-\sqrt{(20-x)^2+16}$ by F and G respectively, where $0<x<20$. Let P be a moving point on F. The vertical line passing through P cuts G at the point Q. Denote the x-coordinate of P by u. It is given that the length of PQ attains its minimum value when $u=a$.

(a) Find a. (4 marks)

(b) The horizontal line passing through P cuts the y-axis at the point R while the horizontal line passing through Q cuts the y-axis at the point S.

(i) Someone claims that the area of the rectangle $PQSR$ attains its minimum value when $u=a$. Do you agree? Explain your answer.

(ii) The length of OP increases at a constant rate of 28 units per minute. Find the rate of change of the perimeter of the rectangle $PQSR$ when $u=a$.

(9 marks)

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HKDSE Questions (level 4)

Not for sale

HKCEE Questions (Level 4)

19. *[A.Math 1998 Paper I Q.13]*

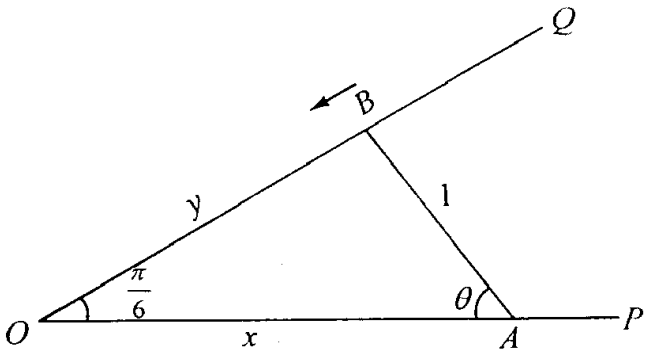


Figure 5

In Figure 5, POQ is a rail and $\angle POQ = \frac{\pi}{6}$. AB is a rod of length 1 m which is free to slide on the rail with end A on OP and end B on OQ. Initially, end A is at the point on OP such that $\angle OAB = \frac{4\pi}{9}$. End B is pushed towards O at a constant speed. After t seconds, $OA = x$ m, $OB = y$ m and $\angle OAB = \theta$, where $0 \le \theta \le \frac{4\pi}{9}$.

(a) Express x and y in terms of θ. (3 marks)

(b) Let S m^2 be the area of ΔOAB.

Show that $\frac{\mathrm{d}S}{\mathrm{d}\theta} = \sin(\frac{5\pi}{6} - 2\theta)$.

Hence find the value of θ such that S is a maximum. (6 marks)

(c) Using (a), show that $\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{-\cos(\frac{5\pi}{6} - \theta)}{\cos\theta}\frac{\mathrm{d}y}{\mathrm{d}t}$. (4 marks)

(d) A student makes the following prediction regarding the motion of end A of the rod :

> As end B moves from its initial position to point O, end A will first move away from O and then it will change its direction and move towards O.

Is the student's prediction correct ? Explain your answer.
(3 marks)

Not for sale

HKCEE Questions (level 4)

Not for sale

HKCEE Questions (Level 5)

20. ***[A.Math 2005 Q.18]***

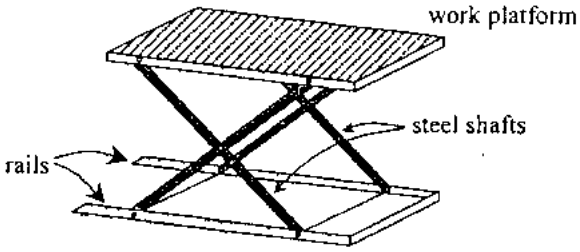


Figure 10

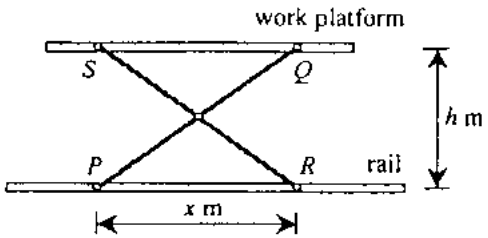


Figure 11

Figure 10 shows an elevating platform for lifting workers to work at different heights. The horizontal work platform is supported by two identical pairs of steel shafts. Figure 11 shows a cross-section of the elevating platform in a vertical plane containing one pair of shafts PQ and RS. The two shafts, each of length 4 m, are hinged at their mid-points. The ends P and R of the shafts can move along a straight horizontal rail with identical uniform speed and in opposite directions. Suppose that the elevating platform is operated under the following conditions:

(*) Initially, PR = 3.6 m. The work platform is then lifted upward by moving the ends P and R of the shafts towards each other such that both PR and SQ decrease at a uniform rate of $\frac{1}{2} ms^{-1}$. Let PR = x m at time t s. It is given that $0.8 \le x \le 3.6$.

(a) Let h m be the height of the work platform above the rail at time t s

(i) Find the range of possible values of h.

(ii) Show that $\frac{dh}{dt} = \frac{x}{2\sqrt{16 - x^2}}$. ***(5 marks)***

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HKCEE Questions (Level 5)

20. ***[A.Math 2005 Q.18] (Cont'd)***

(b) Suppose that the operation of elevating platforms has to comply with the following safety regulation:

"At any instant, the elevating speed of work platforms should not exceed 2 ms^{-1}."

(i) Determine whether the operation of the above elevating platform under the conditions (*) will comply with this regulation.

(ii)

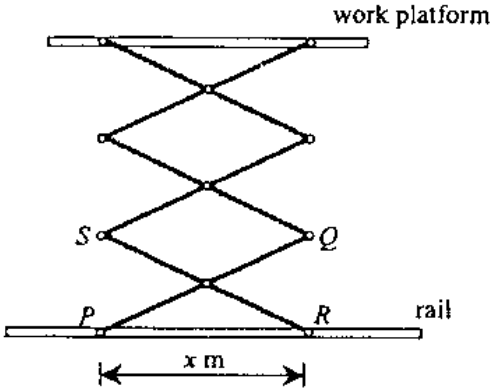


Figure 12 shows a vertical cross-section of a scissors-type elevating platform which can bring workers to a greater height. Two more identical pairs of shafts are installed on each side of the elevating platform as shown. Suppose that this elevating platform is operated under the same conditions (*) as described above. Do you think the operation of this elevating platform will comply with the safety regulation?

If 'Yes', state your reasoning.

If 'No', find the range of possible values of $\frac{dx}{dt}$ in order for the operation of this elevating platform to comply with the safety regulation.

(7 marks)

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More Questions in Past Papers

1. *[A.Math 1990 Paper I Q.11]*

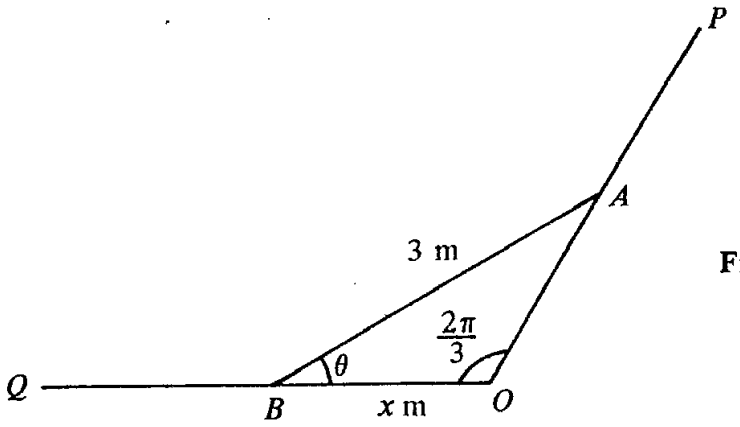


Figure 4

In Figure 4, POQ is a rail where OQ is horizontal and $\angle POQ = \frac{2\pi}{3}$. AB is a rod of length 3 m which is free to slide on the rail with end A on OP and end B on OQ. End A is initially at point O and end B is pushed towards O at a constant speed of $\frac{\sqrt{3}}{3}$ ms^{-1}. After t seconds, B is x metres from O and the rod makes an angle θ with the horizontal.

(a) Express x in terms of θ. (2 marks)

(b) Let S m^2 be the area of ΔAOB.

Show that $\frac{dS}{d\theta} = 3\sqrt{3}\sin(\frac{\pi}{3} - 2\theta)$.

Hence find the maximum value of S. (8 marks)

(c) (i) Show that $\frac{d\theta}{dt} = \frac{1}{6\cos(\frac{\pi}{3} - \theta)}$.

(ii) Find the range of the possible values of $\cos(\frac{\pi}{3} - \theta)$.

Hence determine the greatest and least values of $\frac{d\theta}{dt}$. (6 marks)

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More Questions in Past Papers

2. ***[A.Math 1983 Paper I Q.11]***

Figure 5 shows a rail POQ with $\angle POQ = 120°$. A rod AB of length $\sqrt{7}$ m is free to slide on the rail with its end A on OI' and end B on OQ. Let $OA = x$ m, $OB = y$ m.

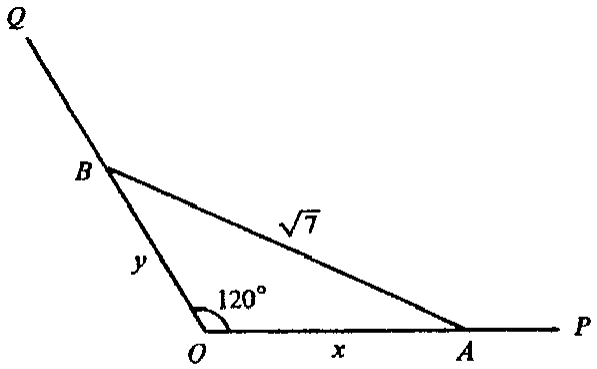


Figure 5

(a) **(i)** Find a relation between x and y .Hence find the value of y when $x = 2$.

(ii) Find $\frac{dy}{dx}$. Given that x and y are functions of time t (in seconds), show that $\frac{dy}{dt} = -\left(\frac{2x+y}{x+2y}\right)\frac{dx}{dt}$. ***(10 marks)***

(b) The end A is pushed towards O with a uniform speed of $\frac{1}{2}$ m/s. When A is at a distance of 2 meters from O, find the speed of the end B. ***(4 marks)***

(c) Suppose the perpendicular distance from O to the rod is p meters. Show that $p = \frac{xy}{2}\sqrt{\frac{3}{7}}$. Hence find $\frac{dp}{dt}$ when $x = 2$. ***(6 marks)***

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More Questions in Past Papers

3. *[A.Math 1994 Paper I Q.12]*

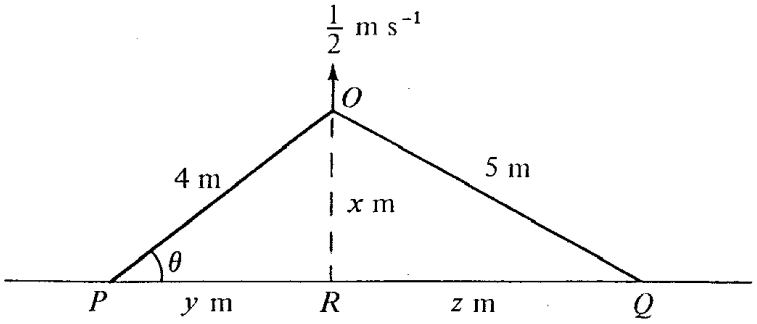


Figure 3

In Figure 3, two rods OP and OQ are hinged at O. The lengths of OP and OQ are 4 m and 5 m respectively. The end O is pushed upwards at a constant rate of $\frac{1}{2}$ m s^{-1} along a fixed vertical axis, and the ends P and Q move along a horizontal rail. R is the projection of O on the rail. At time t seconds, $OR = x$ m and $\angle OPQ = \theta$, where $0 < \theta < \frac{\pi}{2}$.

(a) Express x in terms of θ.

Hence find the rate of change of θ with respect to t in terms of θ.

(3 marks)

(b) Let $PR = y$ m, $RQ = z$ m.

Express $\frac{dy}{dt}$ and $\frac{dz}{dt}$ in terms of θ.

Hence find the rate of change of PQ with respect to t when $\theta = \frac{\pi}{6}$, giving your answer correct to 3 significant figures.

(6 marks)

(c) Find the value of θ such that the area of ΔOPR is a maximum.

By considering the value of $\angle OQR$, find the value of θ such that the area of ΔORQ is a maximum, giving your answer correct to 3 significant figures.

(7 marks)

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4. ***[A.Math 1996 Paper I Q.11]***

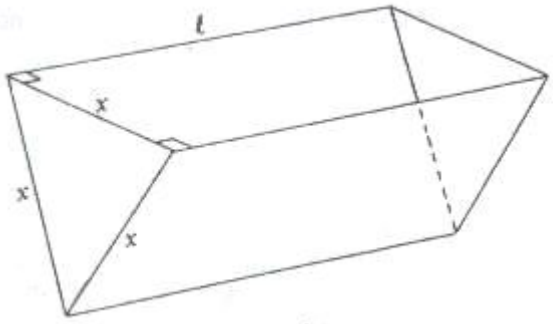


Figure 3(a)



Figure 3(b)

Figure 3(a) shows a vessel with a capacity of 24 cubic units .The length of the vessel is l and its vertical cross- section is an equilateral triangle of side x. the vessel is made of thin metal plates and has no lid. Let S be the total area of metal plates used to make the vessel.

(a) Show that $S = \frac{\sqrt{3}}{2}x^2 + \frac{64\sqrt{3}}{x}$. ***(4 marks)***

(b) Find the values of x and l such that the area of metal plates used to make the vessel is minimum.

(5 marks)

(c) At time t = 0, the vessel described in part (b) is completely filled with water. Suppose the water evaporates at a rate proportional to the area of water surface at that instant such that

$$\frac{dV}{dt} = -\frac{1}{10}A,$$

where V and A and respectively the volume of water and the area of water surface at time t.

(i) Let h be the depth of water in the vessel at time t. (See Figure 3(b)).
Show that $A = 4h$ and $V = 2h^2$
Hence, or otherwise, find $\frac{dh}{dt}$.

(ii) Find the time required for water in the vessel to evaporate completely.

(7 marks)

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More Questions in Past Papers

5. *[A.Math 1988 Paper I Q.12(b)]*

(b)

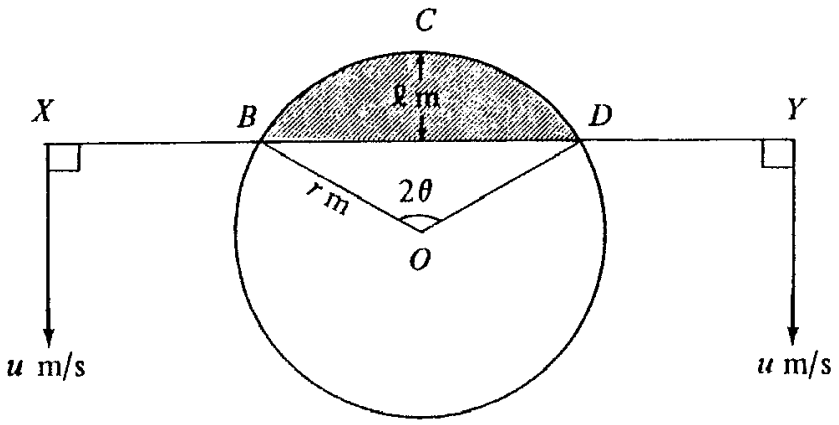


Figure 4

Figure 4 shows a circular pool of radius r metres centred at O. Two men, X and Y, holding the ends of a long rod, are walking in the direction shown at a speed of u metres per second. At a certain instant, the portion BD of the rod subtends an angle of 2θ radians at O and is at a distance ℓ metres from the mid-point C of the rim BD of the pool.

(i) Express ℓ in terms of r and θ.

Let A square metres be the area of the shaded region.

(ii) Express A in terms of r and θ.

(iii) Let $\frac{dA}{dt}$ (in $m^2 s^{-1}$) be the rate of change of the area of the shaded region with respect to time.

Express $\frac{dA}{dt}$ in terms of r, θ and u. (Hint: $\frac{d\ell}{dt} = u$.)

Hence deduce that $\frac{dA}{dt} = ru$ when $\theta = \frac{\pi}{6}$.

(10 marks)

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6. *[A.Math 1991 Paper I Q.12]*

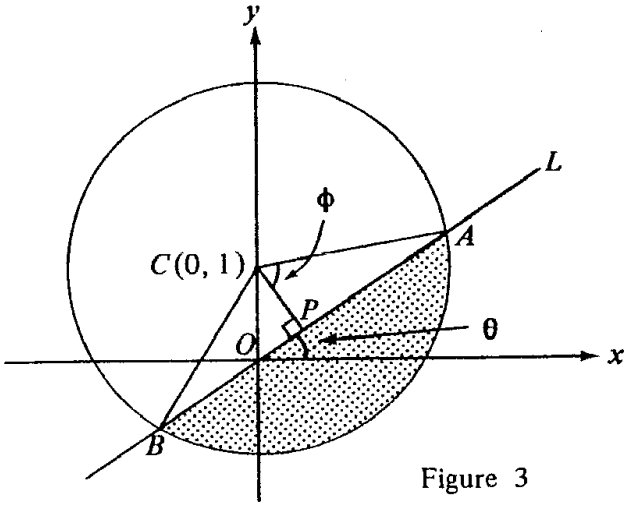


Figure 3

Figure 3 shows a circle of radius 2 centred at the point $C(0, 1)$. A variable straight line L with positive slope passes through the origin O and makes an angle θ with the positive x-axis. L intersects the circle at points A and B. Let S be the area of the shaded segment. P is the point on L such that CP is perpendicular to AB. Let $\angle PCA = \phi$.

(a) (i) Find the length of CP in terms of θ.
Hence show that $\cos\theta = 2\cos\phi$.

(ii) Show that $S = 4\phi - 2\sin 2\phi$.

(5 marks)

(b) (i) Find $\frac{d\phi}{d\theta}$ in terms of θ and ϕ.

(ii) Hence find $\frac{dS}{d\theta}$ in terms of θ.

(7 marks)

(c) L rotates about O in the clockwise direction such that θ decreases steadily at a rate of $\frac{1}{30}$ radian per second. Find the rate of change of S with respect to time when $\theta = \frac{\pi}{3}$.

(4 marks)

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